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BATCH-	16 B.Tech C.S.E

Programming In C-HDOC DAY_11

PROBLEM_1

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum of squares of its digits using a while loop. **E.g.** Number: 123, Sum=14
- Find the sum of cubes of its digits using a for loop. **E.g.** Number: 123, Sum= 36
- Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop. **E.g.** Number: 123, Difference=22

ALGORITHM

1. **Start**
2. Declare the variables choice, num, square, cube, diff, and temp.
3. Ask the user to enter a **positive integer** and store it in num.
4. Display the three choices:
   - Choice 1: Find the sum of squares of the digits.
   - Choice 2: Find the sum of cubes of the digits.
   - Choice 3: Find the difference between the sum of cubes and the sum of squares of the digits.
5. Read the user's choice.
6. Use a **switch statement** based on the choice:
   - **Case 1:**
     1. Set temp = num.
     2. Repeat while temp > 0:
        - Extract the last digit using temp % 10.
        - Add the square of the digit to square.
        - Remove the last digit using temp / 10.
     3. Display square.
   - **Case 2:**
     1. Set temp = num.
     2. Repeat while temp > 0:

            - Extract the last digit using temp % 10.
            - Add the cube of the digit to cube.
            - Remove the last digit using temp / 10.
        3. Display cube.
    - **Case 3:**
        1. Set temp = num.
        2. Repeat while temp > 0:
            - Extract the last digit.
            - Add its square to square.
            - Add its cube to cube.
            - Remove the last digit.
        3. Calculate diff = cube - square.
        4. Display diff.
    - **Default:**
        - Display **"Invalid choice"**.
7. **Stop**.

C PROGRAM CODE

```
#include <stdio.h>
#include <math.h>
int main()
{
  int choice,num,square=0,cube=0,diff=0,temp;
  printf("enter a positive integer number\n");
  scanf("%d", &num);
  printf(" Choice 1 is to find the sum of square of its digits\n");
  printf("choice 2 is to find the cube of the digits of the numbers\n");
  printf("Choice 3 is to find the diff between the cube and square of its digits\n");
  printf("enter your choice:\n");
  scanf("%d", &choice);
  switch(choice)
  {
    case 1:
     temp=num;
```

```
        while(temp>0)
        {
         int digit=temp%10;
         square=square+(int)pow(digit,2);
         temp=temp/10;
        }
        printf("the sum of the square of the digits is %d\n",square);
        break;
      case 2:

         for(temp=num;temp>0;temp=temp/10)
         {
            int digit=temp%10;
            cube=cube+(int)pow(digit,3);
         }
         printf("the sum of the cube of the digits is %d\n",cube);
         break;
      case 3:
         for(temp=num;temp>0;temp=temp/10)
         {
            int digit=temp%10;
            square=square+(int)pow(digit,2);
            cube=cube+(int)pow(digit,3);
         }
         diff=cube-square;
         printf("the diff between the cube and the square of the digits is %d\n",diff);
         break;
      default:
         printf("invalid choice\n");
   }
   return 0;
}
```

HDOC_DAY11 REPORT

TEST CASES

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	enter a positive integer number 123 Choice 1 is to find the sum of square of its digits choice 2 is to find the cube of the digits of the numbers Choice 3 is to find the diff between the cube and square of its digits enter your choice: 1	the sum of the square of the digits is 14	the sum of the square of the digits is 14	PASS
2	enter a positive integer number 343 Choice 1 is to find the sum of square of its digits choice 2 is to find the cube of the digits of the numbers Choice 3 is to find the diff between the cube and square of its digits enter your choice: 2	the sum of the cube of the digits is 118	the sum of the cube of the digits is 118	PASS